Junhee Park

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EDUCATION

University of Toronto — Toronto, ON
B.A.Sc. Engineering Science - Aerospace Engineering Major + PEY — Expected 2029
- CGPA: 3.86/4.00; Dean's Honour List (all terms); $15,000 in merit scholarships, incl. Anthony A. Haasz Scholarship.

RESEARCH AND ENGINEERING EXPERIENCE

University of Toronto Institute for Aerospace Studies (UTIAS) — Toronto, ON
Undergraduate Researcher, Prof. S. Chaudhuri | Pratt & Whitney-sponsored research — May 2026-Present
- Develop and validate compressible OpenFOAM models of contrail-relevant aircraft exhaust mixing using RANS/URANS and LES, including variable-density thermal flow and water-vapour transport.
- Validated an axisymmetric compressible-flow model against contrail-tunnel temperature measurements, achieving 1.42% MAPE across six stations (2.64% max) versus over 40% for the planar baseline.
- Designed a Python-generated, RANS-informed structured O-grid with 14.3 million cells for LES; automated OpenFOAM simulations on Trillium using Bash, Slurm, and OpenMPI, scaling to 576 MPI ranks.
- Ran a 9-case bypass-flow study spanning bypass ratios of 2.5-10 and temperatures of 240-300 K; quantified cooling, water-vapour dilution, and ice supersaturation, with colder bypass flow shifting onset upstream by up to 30%.
- Diagnosed coupled velocity/thermal-density instabilities in compressible LES through controlled solver, mesh, discretization, and turbulence-model studies; established a stable density-based LES framework and quantified unsteady supersaturation using volume-weighted RMS, percentile, and intermittency statistics.

University of Toronto Aerospace Team - Unmanned Aerial Systems SAE — Toronto, ON
Aerodynamics and Aircraft Design Engineer — September 2025-Present
- Contribute to aerodynamic design, manufacturing, integration, and flight testing of an autonomous hybrid VTOL for SAE Aero Design Advanced; team placed 4th in Flight and 3rd in Design Report at the 2026 international competition in Texas.
- Built OpenVSP/VSPAero models to evaluate propwash, spanwise loading, induced drag, and wing-propeller interaction; resolved geometry, paneling, wake, and solver issues and cross-validated trends using FLOW5, XFLR5, and lifting-line methods.
- Extended a MATLAB multidisciplinary design optimization framework with fuselage drag and mass models; performed design-space exploration using surrogate and pattern-search optimization of wing, fuselage, and spar geometry under aerodynamic, structural, takeoff, climb, stability, and power constraints.
- Designed wing assembly jigs and manufactured balsa and 3D-printed wing concepts using SolidWorks and laser-cut tooling; supported aircraft integration and flight testing, including successful runway takeoff and landing.

Robotics for Space Exploration — Toronto, ON
Rover Arm Robotics Engineer — September 2024-August 2025
- Redesigned a rover-arm elbow transmission to reduce component mass by approximately 30% while maintaining torque capacity and reducing backlash using a hex-hub locking mechanism; designed and prototyped an end effector and modular test fixture in Fusion 360 using machining and 3D printing.

SELECTED TECHNICAL PROJECTS

Desktop Wind Tunnel and Vortex-Shedding Analysis — Independent Project
- Designed and 3D-printed a desktop wind tunnel; developed a Python/OpenCV workflow to extract vortex-shedding frequency from video and compare the measured Strouhal number with theoretical predictions.

Numerical Methods and Physics-Informed Neural Networks — Computational Research Paper
- Implemented Euler, improved Euler, and Runge-Kutta solvers and trained a physics-informed neural network for the nonlinear Brusselator system; benchmarked convergence, stability, computational cost, and phase-space behaviour.

TECHNICAL SKILLS

CFD and Aerodynamics: OpenFOAM, ParaView, OpenVSP/VSPAero, FLOW5, XFLR5; RANS, URANS, LES, compressible flow, multi-species transport, meshing, multidisciplinary design optimization (MDO), design-space exploration
Programming and HPC: Python, MATLAB, C/C++, Bash; Linux, Git, Slurm, OpenMPI, Trillium HPC, OpenCV
CAD and Manufacturing: SolidWorks, Fusion 360, Onshape; 3D printing, laser cutting, machining, balsa construction

LEADERSHIP AND MUSIC

Lead violinist, Engineering Drama Society (Come From Away, 2026); performer, Engineering Science Nocturne (2025, 2026); former concertmaster, LYS (2019-2024); former concerto soloist, LCO and LCSS (2021-2024).